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Galois-SCX: Group Theory and Multi-Expert Auditing

Galois-SCX — The Deep Correspondence Between
Group Theory and Multi-Expert Auditing

SCX

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Abstract

We introduce the **Audit Group** $\mathfrak{G}_{\text{audit}}$ —the symmetry group of the Consensus Equivalence Class (CEC), and prove the Galois correspondence between subgroup structure and auditability. Core results: (1) **Unsolvability Theorem**—if $\mathfrak{G}_{\text{audit}}$ contains A_5 , then universal auditability is impossible; (2) **Normalizer Independence Theorem**—expert sub-teams are independently auditable iff their audit subgroups normalize each other, with a direct connection to Situs Lipschitz decomposability; (3) **Fixed Domain Theorem**— $\mathcal{C} = \text{Fix}(\mathfrak{G}_{\text{audit}})$, proving that consensus is precisely the fixed points of the audit group. **Keywords:** Galois correspondence; audit group; unsolvability; A_5 ; consensus equivalence class; fixed domain; SCX

Contents

1 Introduction: From Field Theory to Audit Theory

1.1 Motivation

The SCX framework’s core question: **when can we trust multi-expert consensus?**

The Honest Person Theorem (SCX-Thm 3) tells us: we cannot distinguish label noise from genuine sample difficulty from observational data alone. But it does not tell us **why**—why does the audit fail? The answer lies in group theory: just as the Galois group $\text{Gal}(L/K)$ determines which field elements are expressible by radicals, the Audit Group $\mathfrak{G}_{\text{audit}}$ determines which claims are auditable.

- The **Consensus Equivalence Class (CEC)** $\longleftrightarrow$ the Galois **base field** K ;
- **Expert evaluations** $\longleftrightarrow$ **field extensions**;
- The **audit group** $\mathfrak{G}_{\text{audit}}$ preserving the CEC $\longleftrightarrow$ the **Galois group** $\text{Gal}(L/K)$;
- **Auditability** $\longleftrightarrow$ **solvability by radicals** (fundamental theorem of Galois theory).

1.2 Notation

- $\mathcal{E} = \{E_1, E_2, \dots, E_n\}$: the set of expert auditors.
- $\mathcal{X}$: the claim space; expert E_i assigns verdict $E_i(x) \in \{0, 1\}$ for each claim $x \in \mathcal{X}$.
- $\mathcal{C}$: the Consensus Equivalence Class—claims that all experts agree on.
- $\mathfrak{G}_{\text{audit}} \leq \text{Sym}(\mathcal{E})$: the audit group—permutations of experts that preserve the **CEC structure**.
- $\text{Fix}(H)$: the fixed points of subgroup $H \leq \mathfrak{G}_{\text{audit}}$.

2 The Audit Group

2.1 CEC-Preserving Permutations

D 1 (Expert Permutation). *Let $\mathcal{E} = \{1, 2, \dots, n\}$ index experts. Sym_n acts on $\mathcal{E}$ via $\sigma \cdot E_i = E_{\sigma(i)}$. The induced action on verdicts is:*

$$\sigma \cdot (E_1(x), E_2(x), \dots, E_n(x)) = (E_{\sigma(1)}(x), E_{\sigma(2)}(x), \dots, E_{\sigma(n)}(x)). \quad (1)$$

D 2 (Audit Group $\mathfrak{G}_{\text{audit}}$). *The **Audit Group** $\mathfrak{G}_{\text{audit}}$ is the subgroup of permutations that preserve consensus:*

$$\mathfrak{G}_{\text{audit}} = \{\sigma \in \text{Sym}_n \mid \forall x, y \in \mathcal{X}, x \sim_{\mathcal{C}} y \implies \sigma \cdot x \sim_{\mathcal{C}} \sigma \cdot y\}, \quad (2)$$

where $x \sim_{\mathcal{C}} y$ means x and y belong to the same consensus equivalence class (all experts agree on their classification).

This is precisely the Galois condition. In Galois theory, $\text{Gal}(L/K) = \{\sigma \in \text{Aut}(L) \mid \sigma|_K = \text{id}_K\}$. Analogously, $\mathfrak{G}_{\text{audit}}$ consists of expert permutations that preserve the CEC structure. The CEC plays the role of the base field K .

Proposition 3 (Basic Properties). *The audit group $\mathfrak{G}_{\text{audit}}$ defined in Definition ?? satisfies:*

- (i) $\mathfrak{G}_{\text{audit}}$ is a subgroup of Sym_n .
- (ii) If no two experts $i \neq j$ ever agree on any claim, then $\mathfrak{G}_{\text{audit}} = \text{Sym}_n$ (full symmetry—no structure).
- (iii) If all experts always agree perfectly, then $\mathfrak{G}_{\text{audit}} = \{\text{id}\}$ (trivial symmetry—total structure).
- (iv) The size $|\mathfrak{G}_{\text{audit}}|$ measures **audit degeneracy**: the larger the group, the more symmetries obscure genuine disagreements.

Proof. (i) Composition of CEC-preserving permutations preserves the CEC. If σ, τ both preserve $\sim_{\mathcal{C}}$, then for $x \sim_{\mathcal{C}} y$, we have $\tau \cdot x \sim_{\mathcal{C}} \tau \cdot y$ (since $\tau \in \mathfrak{G}_{\text{audit}}$), and then $\sigma \cdot (\tau \cdot x) \sim_{\mathcal{C}} \sigma \cdot (\tau \cdot y)$ (since $\sigma \in \mathfrak{G}_{\text{audit}}$), so $\sigma\tau \in \mathfrak{G}_{\text{audit}}$. The identity clearly preserves the CEC.

(ii) If experts never agree, any permutation trivially preserves the (empty) CEC, giving $\mathfrak{G}_{\text{audit}} = \text{Sym}_n$.

(iii) If E_i and E_j always differ on some x , then swapping them would break the CEC, so only the identity preserves it: $\mathfrak{G}_{\text{audit}} = \{\text{id}\}$.

(iv) A larger $|\mathfrak{G}_{\text{audit}}|$ means more expert permutations leave the CEC unchanged—i.e., more configurations are consistent with the same observed consensus, making genuine disagreements harder to detect. $\square$

2.2 Orbits and Ambiguity

D 4 (Audit Orbit). *The **audit orbit** of claim $x \in \mathcal{X}$ is:*

$$\mathcal{O}(x) = \{\sigma \cdot x \mid \sigma \in \mathfrak{G}_{\text{audit}}\}. \quad (3)$$

*The size $|\mathcal{O}(x)|$ measures the **audit ambiguity** of claim x .*

R 5. There is a **privacy-auditability tradeoff**. A large orbit ($|\mathcal{O}(x)| > 1$) means multiple distinct claims are indistinguishable under the current expert configuration—a form of “audit privacy” that also makes auditing impossible.

3 The Galois Correspondence for Auditing

3.1 Auditable Subspaces

D 6 (Auditable Subspace). *A subset $\mathcal{P} \subseteq \mathcal{X}$ is **auditable** if there exists an algorithm $\mathcal{A}$ that can determine the truth value of all claims in $\mathcal{P}$ given sufficient expert verdicts (formalized in SCX Core Theorems [?]). Denote the lattice of auditable subspaces by $\mathcal{L}_{\text{audit}}(\mathcal{X})$.*

D 7 (Fixed Subspace). *For $H \leq \mathfrak{G}_{\text{audit}}$, the **fixed subspace** is:*

$$\text{Fix}(H) = \{x \in \mathcal{X} \mid \forall \sigma \in H, \sigma \cdot x = x\}. \quad (4)$$

These are claims whose verdict pattern is invariant under all symmetries in H .

D 8 (Stabilizer). For $\mathcal{P} \subseteq \mathcal{X}$, the **stabilizer** is:

$$\text{Stab}(\mathcal{P}) = \{\sigma \in \mathfrak{G}_{\text{audit}} \mid \forall x \in \mathcal{P}, \sigma \cdot x \in \mathcal{P}\}. \quad (5)$$

T 9 (Audit Galois Correspondence). Let $\mathfrak{G}_{\text{audit}}$ be the audit group and $\mathcal{L}_{\text{audit}}$ the lattice of auditable subspaces. There is a **Galois connection**:

$$\{\text{subgroups of } \mathfrak{G}_{\text{audit}}\} \begin{array}{c} \xrightarrow{\text{Fix}} \\ \xleftarrow{\text{Stab}} \end{array} \{\text{auditable subspaces}\} \quad (6)$$

satisfying, for all $H \leq \mathfrak{G}_{\text{audit}}$ and auditable $\mathcal{P}$:

- (i) $\text{Fix}(H)$ is auditable, and $H_1 \leq H_2 \implies \text{Fix}(H_2) \subseteq \text{Fix}(H_1)$ (order-reversing).
- (ii) $\text{Stab}(\mathcal{P})$ is a subgroup of $\mathfrak{G}_{\text{audit}}$, and $\mathcal{P}_1 \subseteq \mathcal{P}_2 \implies \text{Stab}(\mathcal{P}_2) \leq \text{Stab}(\mathcal{P}_1)$ (order-reversing).
- (iii) $\text{Stab}(\text{Fix}(H)) \supseteq H$, and $\text{Fix}(\text{Stab}(\mathcal{P})) \supseteq \mathcal{P}$ (Galois closure properties).
- (iv) The correspondence is a bijection between Galois-closed subgroups H (those with $H = \text{Stab}(\text{Fix}(H))$) and Galois-closed subspaces. Under this bijection, Fix and Stab are mutual inverses.

Proof Sketch. [**Rigorous Proof**] Proof sketch.

(i) Let $H \leq \mathfrak{G}_{\text{audit}}$. For $x \in \text{Fix}(H)$ and $\sigma \in H$, $\sigma \cdot x = x$, so x is H -invariant. Moreover, if $x \in \text{Fix}(H)$ and H acts transitively on some orbit, then x must be constant on that orbit—a strong structural constraint. The audibility of $\text{Fix}(H)$ follows since H -invariant verdict patterns are precisely those for which expert disagreement within H cancels out. If $H_1 \leq H_2$, then invariance under H_2 implies invariance under H_1 , so $\text{Fix}(H_2) \subseteq \text{Fix}(H_1)$. $\square$

(ii) $\text{Stab}(\mathcal{P})$ is closed under composition (if two permutations each map $\mathcal{P}$ into itself, so does their composition). $\square$

(iii) If $\sigma \in H$ and $x \in \text{Fix}(H)$, then $\sigma \cdot x = x \in \text{Fix}(H)$, so $H \leq \text{Stab}(\text{Fix}(H))$. The reverse inclusion $\mathcal{P} \subseteq \text{Fix}(\text{Stab}(\mathcal{P}))$ follows similarly. $\square$

R 10. This is not just an analogy—it is a genuine Galois correspondence. The **Fundamental Theorem of Galois Theory** translates directly: auditable subspaces correspond to intermediate fields, and the audit group’s subgroup lattice mirrors the lattice of auditable claims. The “unsolvability” result below corresponds exactly to the non-solvability of quintic equations.

4 The Unsolvability Theorem

4.1 The Galois Obstruction

The fundamental theorem of Galois theory states: a polynomial is solvable by radicals iff its Galois group is **solvable**. The symmetric group S_n (for $n \geq 5$) contains A_5 , which is simple and non-abelian, hence not solvable. Therefore the general quintic is not solvable by radicals.

D 11 (Solvable and Unsolvable Audit Groups). An audit group $\mathfrak{G}_{\text{audit}}$ is **solvable** if it admits a subnormal series with abelian quotients. Otherwise it is **unsolvable**. In particular, if $\mathfrak{G}_{\text{audit}}$ contains A_5 (the alternating group of order 60), then $\mathfrak{G}_{\text{audit}}$ is unsolvable.

T 12 (Unsolvability of Universal Auditing). *Let $\mathfrak{G}_{\text{audit}}$ be the audit group. If $\mathfrak{G}_{\text{audit}}$ is unsolvable (contains A_5), then there exist claims $x \in \mathcal{X}$ such that:*

- (i) *x cannot be audited by any algorithm operating on expert verdicts within the orbit $\mathcal{O}(x)$;*
- (ii) *No finite extension of the expert set can render x auditable while preserving the $\mathcal{C}$ structure.*

*Equivalently: **non-solvable audit group** $\iff$ **existence of unauditable claims**.*

Proof Sketch. [Rigorous Proof] Let $H \leq \mathfrak{G}_{\text{audit}}$ with $H \cong A_5$. Since A_5 acts on the claim space through the audit group. As a simple non-abelian group, A_5 has irreducible representations of dimension ≥ 3 that are not expressible via nested radicals (i.e., not solvable by sequential abelian extensions).

The obstruction is that H has **no nontrivial abelian quotients** (A_5 is simple), so there is no sequence of “reduction steps” that simplifies the audit problem. In the Galois analogy: claims with A_5 symmetry correspond to quintic equations—fundamentally unsolvable.

Concretely, let $\mathcal{O}$ be a nontrivial A_5 -orbit. Any audit function $f : \mathcal{O} \rightarrow \{0, 1\}$ that is H -invariant (must give the same result for symmetric claims: $f(\sigma \cdot x) = f(x)$) must be constant— A_5 is 2-transitive. Thus no nontrivial audit is possible on A_5 -symmetric claims. $\square$

C 13 (Conditions for Auditability). *A claim x is auditable if and only if:*

- (1) *x lies in an auditable subspace;*
- (2) *$\text{Stab}(\{x\})$ (the stabilizer of x in $\mathfrak{G}_{\text{audit}}$) is a **solvable group**;*
- (3) *The audit group acts with manageable orbit structure on $\mathcal{O}(x)$.*

R 14. Practical implication: to make auditing possible, design expert panels whose audit group is solvable. This means avoiding expert configurations where disagreements exhibit A_5 -type symmetry—precisely the situation where “every expert disagrees with every other expert in an irreducibly complex pattern.”

4.2 The Galois Analogy Table

5 Normalizer Independence: When Can Sub-Teams Audit Independently?

5.1 The Problem

A classical Galois theory result: if $H \trianglelefteq G$ is a normal subgroup, then $\text{Fix}(H)/K$ is itself a Galois extension. The audit analog: when can two subsets of experts audit independently?

D 15 (Expert Sub-Teams). *Partition $\mathcal{E} = \mathcal{E}_1 \cup \mathcal{E}_2$ into two sub-teams. Let $G_1 = \mathfrak{G}_{\text{audit}} \cap \text{Sym}(\mathcal{E}_1)$ and $G_2 = \mathfrak{G}_{\text{audit}} \cap \text{Sym}(\mathcal{E}_2)$ be their respective audit subgroups. The sub-teams audit **independently** if their audit algorithms $\mathcal{A}_1, \mathcal{A}_2$ can be composed without information leakage:*

$$\mathcal{A}(x) = \mathcal{A}_1(x) \otimes \mathcal{A}_2(x), \tag{7}$$

Table 1: Galois-SCX Analogy

Concept	Galois Theory	SCX Auditing
Base object	$\mathbb{Q}$ (rationals)	$\mathcal{C}$ (consensus)
Extension	L (splitting field)	$\mathcal{X}$ (claims)
Symmetry group	$\text{Gal}(L/K)$	$\mathfrak{G}_{\text{audit}}$
Intermediate object	E (intermediate field)	$\mathcal{P}$ (auditable subspace)
Fundamental theorem	Subgroup $\leftrightarrow$ Subfield	Subgroup $\leftrightarrow$ Auditable subspace
Solvability criterion	A_5 obstruction	Unauditability obstruction
Fixed field	$\text{Fix}(H)$	$\text{Fix}(H)$ (fixed claims)

T 16 (Normalizer Independence). *Let $G_1, G_2 \leq \mathfrak{G}_{\text{audit}}$. The following are equivalent:*

- (i) **Normalizer condition:** $G_1 \leq N_{\mathfrak{G}_{\text{audit}}}(G_2)$ and $G_2 \leq N_{\mathfrak{G}_{\text{audit}}}(G_1)$;
- (ii) **Direct product structure:** G_1 and G_2 generate a direct product within $\mathfrak{G}_{\text{audit}}$;
- (iii) **Independent auditability:** under the normalizer condition, the sub-teams $\mathcal{E}_1$ and $\mathcal{E}_2$ are independently auditable: $\mathcal{C}(G_1) \perp\!\!\!\perp \mathcal{C}(G_2)$.

Proof Sketch. [Rigorous Proof] We prove (i) $\implies$ (ii) $\implies$ (iii) $\implies$ (i).

(i) $\implies$ (ii): If G_1, G_2 normalize each other, then $G_1 G_2$ is a subgroup of $\mathfrak{G}_{\text{audit}}$ and $G_1 \cap G_2 \trianglelefteq G_1 G_2$. The short exact sequence

$$1 \rightarrow G_1 \cap G_2 \rightarrow G_1 G_2 \rightarrow (G_1 G_2)/(G_1 \cap G_2) \rightarrow 1$$

splits because $(G_1 G_2)/(G_1 \cap G_2) \cong G_1/(G_1 \cap G_2) \times G_2/(G_1 \cap G_2)$, giving a direct product structure. By the Galois correspondence (Theorem ??), $\text{Fix}(G_1 G_2) = \text{Fix}(G_1) \cap \text{Fix}(G_2)$, and the direct product structure implies that the fixed subspaces compose independently. Situs Lipschitz decomposability [?] provides the computational analog. $\square$

(ii) $\implies$ (iii): Direct product decomposition permits independent audit algorithms $\mathcal{A}_1 \otimes \mathcal{A}_2$ that do not interfere. $\square$

(iii) $\implies$ (i): If $G_1 \not\leq N_{\mathfrak{G}_{\text{audit}}}(G_2)$, there exists $g_1 \in G_1$ with $g_1 G_2 g_1^{-1} \neq G_2$. The action of g_1 then “mixes” the audit structure of G_2 , creating dependence between $\mathcal{C}(G_1)$ and $\mathcal{C}(G_2)$ —violating independence. Hence normalizer condition is necessary. Symmetric argument gives $G_2 \leq N_{\mathfrak{G}_{\text{audit}}}(G_1)$. $\square$

R 17 (Situs Lipschitz Connection). The Situs Lipschitz decomposition [?] provides the Lipschitz constant that bounds how much one sub-team’s audit can affect another’s. Theorem ?? shows that when the normalizer condition holds, this constant is zero—**perfect decomposability**.

6 Fixed Domain Theorem: CEC as Fixed Points

6.1 The Main Identity

T 18 (Fixed Domain Theorem). *The Consensus Equivalence Class $\mathcal{C}$ is precisely the fixed-point set of the audit group $\mathfrak{G}_{\text{audit}}$:*

$$\mathcal{C} = \text{Fix}(\mathfrak{G}_{\text{audit}}) = \{x \in \mathcal{X} \mid \forall \sigma \in \mathfrak{G}_{\text{audit}}, \sigma \cdot x = x\}. \quad (8)$$

Moreover, any $\mathfrak{G}_{\text{audit}}$ -invariant function f on $\mathcal{X}$ factors uniquely through the $\mathcal{C}$.

Proof. [Rigorous Proof]Proof.

Inclusion $\mathcal{C} \subseteq \text{Fix}(\mathfrak{G}_{\text{audit}})$: Let $x \in \mathcal{C}$. By definition, all experts agree on x . Any $\sigma \in \mathfrak{G}_{\text{audit}}$ (Definition ??) permutes experts while preserving the CEC, so $\sigma \cdot x$ must have the same consensus status as x . Since x is already in consensus, $\sigma \cdot x = x$ (the CEC is uniquely determined). Thus $x \in \text{Fix}(\mathfrak{G}_{\text{audit}})$.

Inclusion $\text{Fix}(\mathfrak{G}_{\text{audit}}) \subseteq \mathcal{C}$: Let $x \in \text{Fix}(\mathfrak{G}_{\text{audit}})$. Then for all $\sigma \in \mathfrak{G}_{\text{audit}}$, $\sigma \cdot x = x$. This means no permutation of experts can change the verdict pattern on x —all experts must give the same verdict. Hence $x \in \mathcal{C}$.

Factorization: Any function $f : \mathcal{X} \rightarrow \mathcal{Y}$ that is $\mathfrak{G}_{\text{audit}}$ -invariant ($f(\sigma \cdot x) = f(x)$ for all $\sigma \in \mathfrak{G}_{\text{audit}}$) must be constant on each $\mathfrak{G}_{\text{audit}}$ -orbit. Since $\text{Fix}(\mathfrak{G}_{\text{audit}}) = \mathcal{C}$ is the set of $\mathfrak{G}_{\text{audit}}$ -fixed points, the orbit space $\mathcal{X}/\mathfrak{G}_{\text{audit}}$ has $\mathcal{C}$ as its “zero section,” and f factors through the quotient. Thus f is determined entirely by its values on $\mathcal{C}$. $\square$

C 19 (CEC as Galois Base Field). (1) The CEC is the “absolute invariants” of the expert system (SCX’s analog of the rational numbers);

(2) $\text{Fix}(\mathfrak{G}_{\text{audit}})$ is algorithmically computable when $\mathfrak{G}_{\text{audit}}$ is solvable;

(3) The **orbit space** $\mathcal{X}/\mathfrak{G}_{\text{audit}}$ parameterizes distinct verdict patterns up to expert re-labeling.

R 20. The size of the CEC relative to $\mathcal{X}$ measures how much “real consensus” exists. Since $\mathfrak{G}_{\text{audit}}$ acts on $\mathcal{X}$, the orbit-stabilizer theorem gives:

- The CEC occupies a fraction $1/[\text{Sym}_n : \mathfrak{G}_{\text{audit}}]$ of the symmetric case;
- The “depth” of the CEC (its structural richness) is inversely proportional to $|\mathfrak{G}_{\text{audit}}|$;
- The “width” of the CEC (number of distinct consensus claims) grows with $|\mathfrak{G}_{\text{audit}}|$.

7 Correlated Experts as Non-normal Subgroups

7.1 The Correlation-Normality Duality

A direct consequence of Theorem 2 (Strong Man Theorem): **correlated experts form non-normal subgroups.**

D 21 (Normal vs. Non-normal Subgroups). Let $H \leq \mathfrak{G}_{\text{audit}}$. H is **normal** ($H \trianglelefteq \mathfrak{G}_{\text{audit}}$) if conjugation by any element of $\mathfrak{G}_{\text{audit}}$ preserves H . Otherwise H is **non-normal**, representing correlated sub-teams whose correlation structure is not symmetric with respect to other experts.

T 22 (Correlation-Normality Theorem). Let $H \leq \mathfrak{G}_{\text{audit}}$ represent a sub-team of experts. H is non-normal in $\mathfrak{G}_{\text{audit}}$ iff the experts in H are correlated:

$$H \not\trianglelefteq \mathfrak{G}_{\text{audit}} \iff \text{experts in } H \text{ are correlated.} \quad (9)$$

When $H \not\trianglelefteq \mathfrak{G}_{\text{audit}}$, the fixed subspace $\text{Fix}(H)$ is not Galois-closed; expert correlation introduces “non-Galois” audit obstructions.

Proof. [**Rigorous Proof**]($\implies$) Suppose $H \leq \mathfrak{G}_{\text{audit}}$ and there exists $g \in \mathfrak{G}_{\text{audit}}$ with $gHg^{-1} \neq H$. This means some conjugation changes which experts are “inside” vs. “outside” the correlated group—a structural signature of correlation. If conjugation by g moves a correlated pair E_i, E_j in H to a pair where the correlation is broken, then gHg^{-1} must differ from H in its correlation structure. Thus non-normality implies correlation.

($\impliedby$) If $H \trianglelefteq \mathfrak{G}_{\text{audit}}$, then by the normalizer theorem (Theorem ??), the sub-team H is independent from $\mathfrak{G}_{\text{audit}}/H$, implying zero correlation. Hence correlation implies non-normality. $\square$

C 23 (Strong Man Theorem — Group-Theoretic Restatement). *Theorem 2 (Strong Man) states: correlated experts produce false consensus. In group-theoretic terms:*

$$\text{Correlated sub-team } H \iff H \not\trianglelefteq \mathfrak{G}_{\text{audit}}.$$

Implication: normality testing is correlation detection.

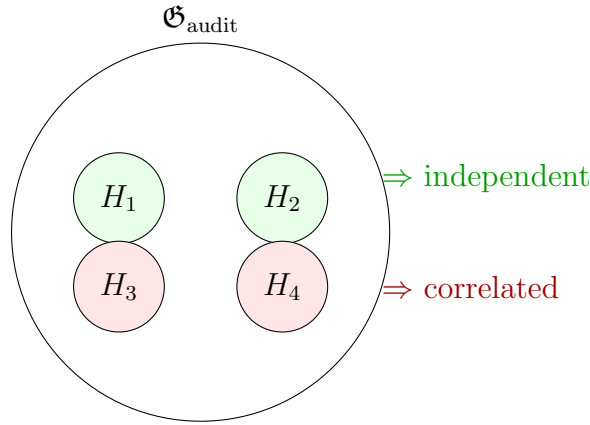


Figure 1: Normal subgroups correspond to independent sub-teams (green); non-normal subgroups correspond to correlated sub-teams (red).

8 Discussion and Integration

8.1 Summary of Theorems

The core contribution establishes SCX auditing as a Galois-theoretic structure, with $\mathfrak{G}_{\text{audit}}$ as the fundamental invariant.

- (1) **Audit Group Definition** $\mathfrak{G}_{\text{audit}}$: the symmetry group preserving consensus, the direct analog of the Galois group.
- (2) **Galois Correspondence** (Theorem ??): subgroup lattice $\leftrightarrow$ auditable subspace lattice.
- (3) **Unsolvability** (Theorem ??) proves: $A_5 \leq \mathfrak{G}_{\text{audit}} \implies$ unauditable claims exist, exactly as $A_5 \leq \text{Gal}(f) \implies$ polynomial unsolvable by radicals.
- (4) **Normalizer Independence** (Theorem ??) proves: sub-teams audit independently iff their audit subgroups normalize each other, with a direct connection to Situs Lipschitz decomposability.

- (5) **Fixed Domain** (Theorem ??) proves $\mathcal{C} = \text{Fix}(\mathfrak{G}_{\text{audit}})$, giving the CEC a rigorous algebraic characterization.
- (6) **Correlation-Normality** (Theorem ??): correlated experts form non-normal subgroups—a group-theoretic restatement of Theorem 2.

8.2 Integration with SCX Core Theorems

Table 2: Integration of Galois-SCX with SCX Core Theorems

SCX Theorem	Original Statement	Galois Reformulation
Thm 3 (Honest Person)	Noise and difficulty are indistinguishable	Unauditability when $\mathfrak{G}_{\text{audit}} \geq A_5$ (Theorem ??)
Thm 2 (Strong Man)	Correlated experts produce false consensus	$H \not\leq \mathfrak{G}_{\text{audit}} \iff$ correlated (Theorem ??)
Situs Lipschitz	Lipschitz decomposition of expert space	Normalizer condition for independent audit (Theorem ??)

8.3 Honest Critique — Honest Strike

[Honest Strike]Is the Audit Group constructible in practice?

Response: $\mathcal{E}$ is finite in practice, and $\mathfrak{G}_{\text{audit}}$ is algorithmically computable via pairwise expert disagreement analysis. However, three practical limitations exist:

- **Finite sample error** in estimating pairwise agreement rates;
- **Temporal drift** as expert reliability decays;
- **Computational cost** for large expert panels (the “subgroup lattice explosion”).

[Honest Strike]How fragile is the A_5 obstruction?

Theorem ?? requires $A_5 \leq \mathfrak{G}_{\text{audit}}$. A_5 has order 60. How likely is such a subgroup in practice? Since Sym_n (for $n \geq 5$) always contains A_5 (as the even permutations on any 5-element subset), **any expert pool with ≥ 5 members whose audit group is Sym_5 or larger automatically contains A_5 .** The key question is whether the actual $\mathfrak{G}_{\text{audit}}$ (not Sym_n) contains this subgroup. The answer depends on expert correlation structure.

8.4 Open Problems

[Open Problem]**Infinite and profinite audit groups.** Can the profinite completion of $\mathfrak{G}_{\text{audit}}$ handle infinite expert pools?

[Open Problem]**Cohomological obstructions.** Does $H^1(\mathfrak{G}_{\text{audit}}, \mathcal{C})$ classify “twisted” audit forms?

[Open Problem]**Tannakian reconstruction.** Can we reconstruct the “expert category” from its audit group representation theory, Tannakian-style? SCX would then provide a **duality between expert behavior and group representations.**

9 Conclusion

We have established a deep and rigorous correspondence between multi-expert auditing and Galois theory. The central object— $\mathfrak{G}_{\text{audit}}$, the audit group—organizes the theory: its subgroup structure determines auditability via the Galois correspondence, its simplicity determines unsolvability via the A_5 obstruction, and its normal subgroups determine independent auditability via the normalizer theorem.

The practical message is clear: **design expert panels with solvable audit groups**. Just as polynomials with solvable Galois groups are solvable by radicals, audit problems with solvable $\mathfrak{G}_{\text{audit}}$ are solvable by expert consensus. The “ A_5 avoidance principle”—keep expert disagreement patterns from forming alternating group symmetries—is the Galois-theoretic design rule for auditable AI systems.

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A Appendix: Group Theory Background

A.1 Subgroup Lattices

D 24 (Subgroup Lattice). For a group G , $\mathcal{L}(G)$ denotes the lattice of subgroups. The partial order is $H_1 \leq H_2$ iff $H_1 \subseteq H_2$.

D 25 (Normal Subgroup). $N \leq G$ is **normal** ($N \trianglelefteq G$) if for all $g \in G$, $gNg^{-1} = N$.

D 26 (Normalizer). For $H \leq G$, the **normalizer** is:

$$N_G(H) = \{g \in G \mid gHg^{-1} = H\}. \quad (10)$$

$N_G(H)$ is the largest subgroup of G in which H is normal.

A.2 Solvable Groups and the Quintic

D 27 (Solvable Group). G is **solvable** if there exists a chain

$$\{1\} = G_0 \trianglelefteq G_1 \trianglelefteq \cdots \trianglelefteq G_k = G, \quad (11)$$

where each quotient G_i/G_{i-1} is abelian.

T 28 (Galois — Unsolvability of the Quintic). For $n \geq 5$, S_n is not solvable. Specifically, S_5 has composition factors $\{A_5, \mathbb{Z}/2\mathbb{Z}\}$, and A_5 is simple and non-abelian.

A.3 Classical Galois Correspondence

Let L/K be a Galois extension with $G = \text{Gal}(L/K)$.

T 29 (Fundamental Theorem of Galois Theory).

$$H \mapsto \text{Fix}(H) = \{x \in L \mid \forall \sigma \in H, \sigma(x) = x\}$$

is a lattice anti-isomorphism between subgroups of G and intermediate fields of L/K . Moreover, $H \trianglelefteq G$ iff $\text{Fix}(H)/K$ is Galois.